

Frequency response. Bode plot and stability

R0005E - Measurement and Feedback Control

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Derivation of the frequency response

We assume that the impulse response $h(t)$ of a linear system is known. Now, we want to describe the system output $y(t)$ for the input $u(t) = \sin \omega t$.

Remember:

- $\sin \omega t = \frac{1}{2j} (e^{j\omega t} - e^{-j\omega t})$
- $u(t) = e^{st} \Rightarrow y(t) = H(s)e^{st}$

From this we can derive

$$y(t) = \frac{1}{2j} [H(j\omega)e^{j\omega t} - H(-j\omega t)e^{-j\omega t}] \quad (1)$$

There, the term $H(-j\omega)$ is the conjugate complex of $H(j\omega)$, which is a complex function. Now we can express the complex function in terms of its absolute value and phase value:

$$\Rightarrow \begin{cases} H(j\omega) &= M(\omega) \cdot e^{j\phi(\omega)} \\ H(-j\omega) &= M(\omega) \cdot e^{-j\phi(\omega)} \end{cases}$$

Now we use these expressions in (1) and rewrite

$$\begin{aligned} \Rightarrow y(t) &= \frac{1}{2j} [M(\omega) \cdot e^{j\phi(\omega)} e^{j\omega t} - M(\omega) \cdot e^{-j\phi(\omega)} e^{-j\omega t}] \\ &= M(\omega) \underbrace{\frac{1}{2j} [e^{j(\omega t - \phi(\omega))} - e^{-j(\omega t + \phi(\omega))}]}_{\text{This is a sine again!}} \\ &= M(\omega) \sin(\omega t + \phi(\omega)) \end{aligned}$$

Conclusions:

- The output $y(t)$ is a sine wave of the same frequency but scaled by $M(\omega)$ and phase shifted by $\phi(\omega)$.
- The frequency response of a system is derived by replacing s by $j\omega$ in the transfer function of the system.

Example:

Let's assume we want to determine the signal amplitude and phase of the output for certain frequencies for the following system:

$$\dot{y} + \frac{1}{40}y = \frac{5}{80}u \quad (2)$$

We know that the input signal is a sine wave at frequency 1 rad/s with amplitude 1.

First we derive the transfer function for (2):

$$H(s) = \frac{2.5}{40s + 1}$$

The frequency response is now given by

$$H(j\omega) = \frac{2.5}{40j\omega + 1}$$

and can be evaluated at $\omega = 1 \text{ rad/s}$ which yields

$$H(j) = \frac{2.5}{40j + 1}$$

This is just a complex number and determines the gain with which the $u(t)$ is amplified and the phase shift which can be observed between input and output.

- **Gain:**

$$|H(j)| = \frac{2.5}{|40j + 1|} = \frac{2.5}{\sqrt{40^2 + 1^2}} \approx \frac{2.5}{40} = 0.0625$$

- **Phase shift:**

$$H(j) = \frac{2.5}{40j + 1} = \frac{2.5}{40j + 1} \cdot \frac{-40j + 1}{-40j + 1} = \frac{2.5 - 100j}{1601}$$

The angle of $H(j)$ is given by the ratio between imaginary and real part:

$$\phi(j) = \arctan \left\{ \frac{\text{Im}(H(j))}{\text{Re}(H(j))} \right\} = \arctan \left\{ \frac{-100}{2.5} \right\} \approx -88^\circ$$

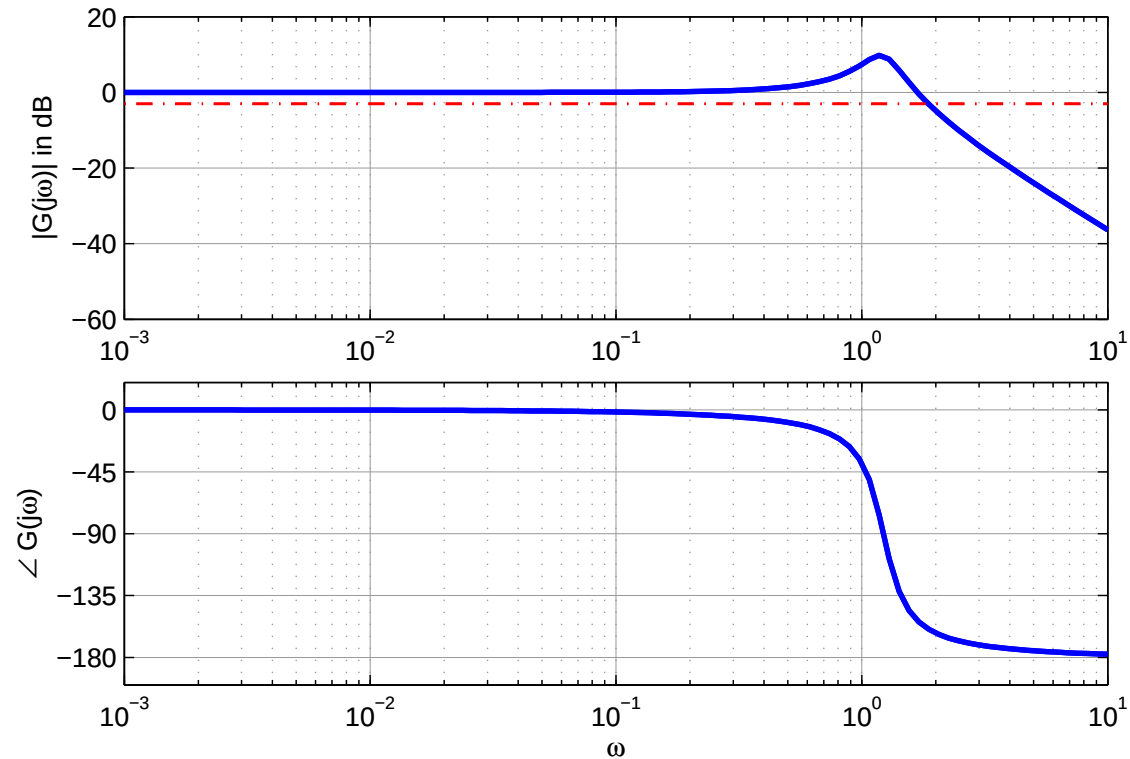
Note:

The denominator will most likely contain complex numbers. By expanding both numerator and denominator with the complex conjugate, you can rewrite the frequency response into a complex number.

Bode plot

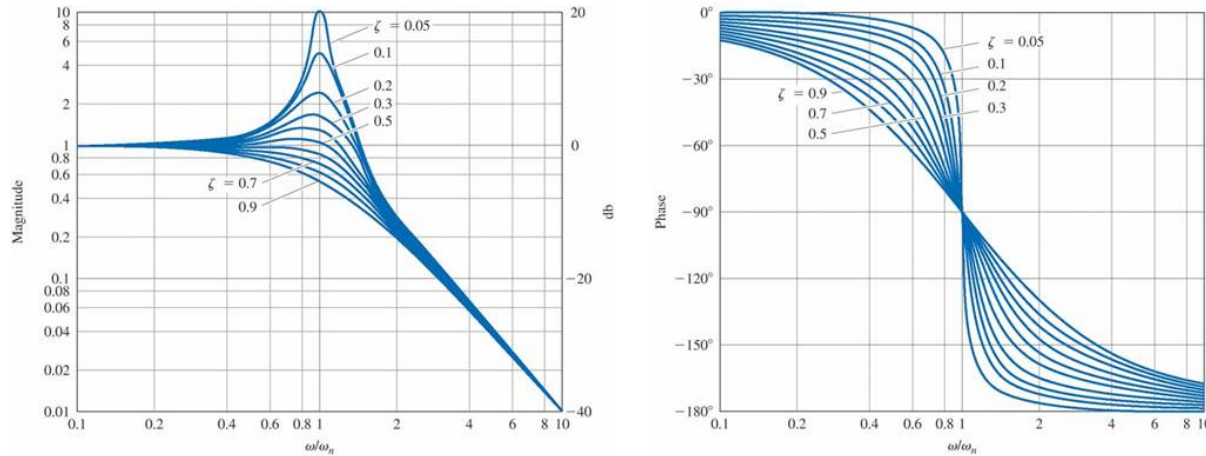
Clearly, we could now generate a plot for the frequency response by plotting the gain and the phase shift over the frequency range $\omega = 0 \dots \infty$.

This yields two plots or diagrams. Usually, the frequency and the magnitude are represented in logarithmic scales.



Note: Conversion between dB and absolute values: $N_{dB} = 20 \log_{10} N_{abs}$

The frequency response can be the response of a closed loop system and not only of the process. Thus, the response can characterize the performance of the closed loop system.



Two important properties:

- **Bandwidth:**

The bandwidth of a system is the maximum frequency at which the output can follow the input in a satisfactory manner.

In practice, this is the maximum frequency at which the magnitude drops below a certain threshold. Usually, $-3dB$ or 0.707 are used as threshold.

- **Resonant peaks:**

Peak in the bodeplot, especially when they exceed 1, are resonant peaks.

The peak level is related to the damping coefficient of a system.

Similar to the root locus, the construction of the bode plot is complicated and standards for typical systems were used to compose the plot. The use of computer programs has then revolutionized this.

In order to construct the plot, the transfer function is rewritten in the factorized form:

$$G(j\omega) = K_0 \frac{\prod_{i=1}^m (j\omega\tau_{Ni} + 1)}{\prod_{i=1}^n (j\omega\tau_{Di} + 1)}$$

This form is also called the Bode form. There the factors are either real values or pairs of complex conjugate values.

If we have a pair of complex conjugate number, then those two terms can be combined to one term. Assume the terms are $(j\omega\tau + 1)$ and $(j\omega\tau^* + 1)$, then

$$\begin{aligned} (j\omega\tau + 1)(j\omega\tau^* + 1) &= j^2\omega^2\tau\tau^* + j\omega(\tau + \tau^*) + 1 \\ &= j^2\omega^2|\tau|^2 + 2j\omega\text{Re}\{\tau\} + 1 \end{aligned}$$

Now we can define: $\omega_n = 1/|\tau|$, $\zeta = \omega\text{Re}\{\tau\}$

$$(j\omega\tau + 1)(j\omega\tau^* + 1) = \left(\frac{j\omega}{\omega_n}\right)^2 + 2\zeta\frac{j\omega}{\omega_n} + 1$$

Thus, this is the standard form for a second order system.

There are three types of terms:

1. $K_0(j\omega)^n$

Factors where the time constant is $\tau \rightarrow \infty$ which corresponds to a root at origo

2. $(j\omega\tau + 1)^{\pm 1}$

First order term with a real root.

3. $\left[\left(\frac{j\omega}{\omega_n} \right)^2 + 2\zeta \frac{j\omega}{\omega_n} + 1 \right]^{\pm 1}$

Second order terms with complex conjugate pair of roots.

The question is now, how can we take advantage of this?

- Since the logarithm is applied to the magnitude of the bode form, it can be shown that the product of the terms resolves into a sum of the logarithm of the individual terms:

$$\begin{aligned} \log_{10} |G(j\omega)| &= \log_{10} \left| K_0 \frac{\prod_{i=1}^m (j\omega\tau_{Ni} + 1)}{\prod_{i=1}^n (j\omega\tau_{Di} + 1)} \right| \\ &= \log_{10} |K_0| + \sum_{i=1}^m \log_{10} |j\omega\tau_{Ni} + 1| - \sum_{i=1}^n \log_{10} |j\omega\tau_{Di} + 1| \end{aligned}$$

- Similarly, the phase of the bode form can be rewritten into the sum of the phases of the individual terms:

$$\begin{aligned}\angle G(j\omega) &= \angle \left\{ K_0 \frac{\prod_{i=1}^m (j\omega\tau_{Ni} + 1)}{\prod_{i=1}^n (j\omega\tau_{Di} + 1)} \right\} \\ &= \angle K_0 + \sum_{i=1}^m \angle \{j\omega\tau_{Ni} + 1\} - \sum_{i=1}^n \angle \{j\omega\tau_{Di} + 1\}\end{aligned}$$

Conclusion:

The Bode plot of a system $G(j\omega)$ is simply derived by adding the contribution of the individual terms for both magnitude and phase.

Type 1: $K_0(j\omega)^n$

- Magnitude:

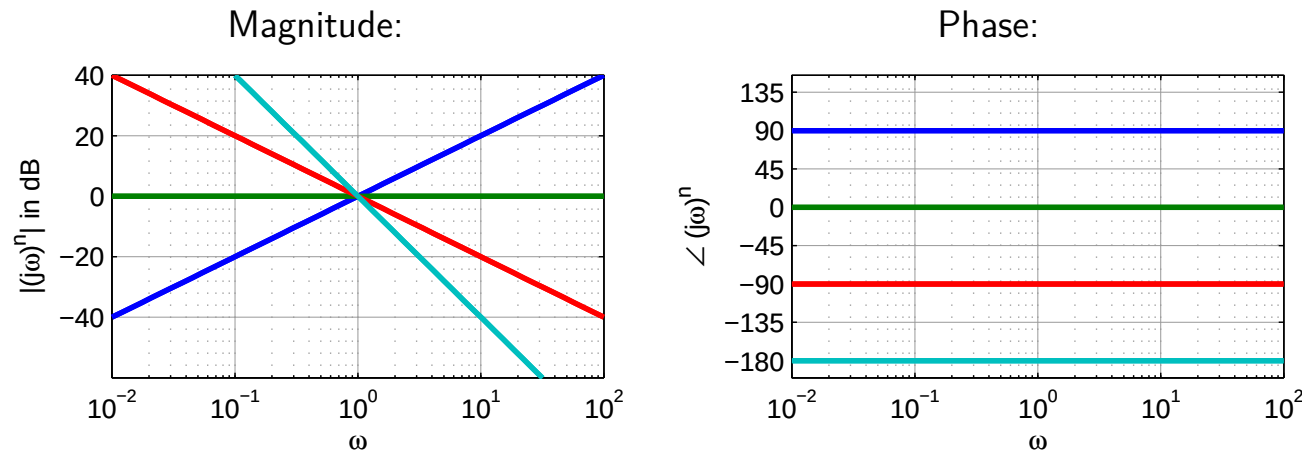
$$\log_{10} |K_0(j\omega)^n| = \log_{10} |K_0| + n \cdot \log_{10} |j\omega| = \log_{10} K_0 + n \cdot \log_{10} \omega$$

For each frequency decade the gain increases with the factor $10n$ with a value for $\omega = 1$ of $\log_{10} |K_0|$.

- Phase:

$$\angle \{K_0(j\omega)^n\} = \angle K_0 + n \cdot \angle(j\omega) = 0 + 90^\circ n$$

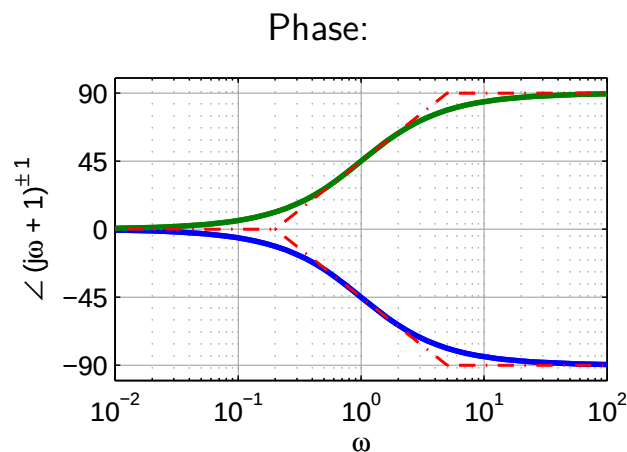
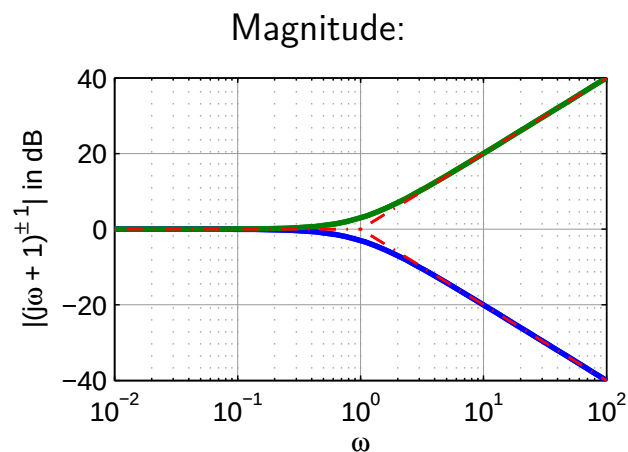
The phase is constant for all frequencies.



$n = 1$: blue, $n = 0$: green, $n = -1$: red, $n = -2$: cyan.

Type 2: $(j\omega\tau + 1)^{\pm 1}$

- $\omega\tau \ll 1$:
For very low frequencies, the term approaches the asymptote 1 for the magnitude and the phase will be 0° . That's the same behavior like the type 1 term with $n = 0$ and $k_0 = 1$.
- $\omega\tau \gg 1$:
For very high frequencies the factor approaches the asymptote $(j\omega\tau)^{\pm 1}$ for the magnitude and the phase approaches $\pm 90^\circ$. This is the same behavior like the type 1 term with $n = \pm 1$ and $K_0 = 1$.
- The point $\omega = 1/\tau$ is called the **break point**. At that point the phase attains the value $\pm 45^\circ$ and the asymptote from $\omega\tau = 0.2$ at 0° to $\omega\tau = 5$ at $\pm 90^\circ$ approximate the change of phase.

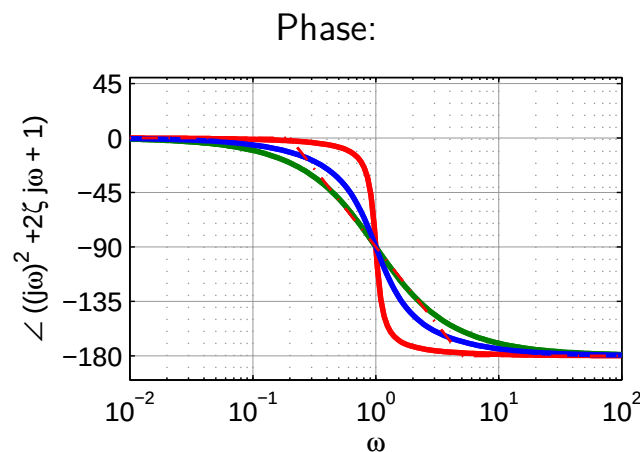
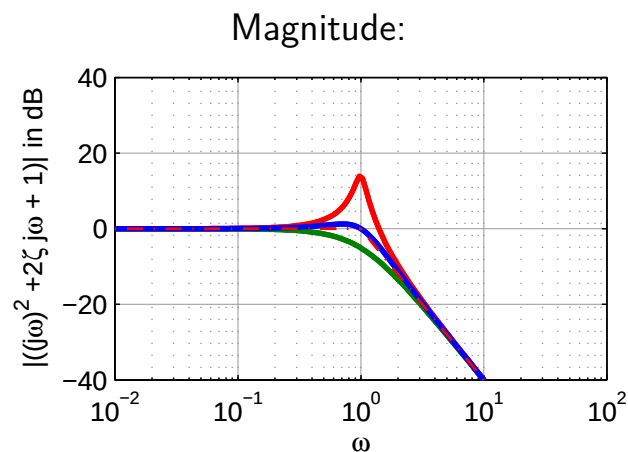


$n = 1$: blue, $n = -1$: green.

Type 3: $\left[\left(\frac{j\omega}{\omega_n} \right)^2 + 2\zeta \frac{j\omega}{\omega_n} + 1 \right]^{\pm 1}$

Principally, the term behaves like the type 2 term with the following differences:

- The break point is at $\omega = \omega_n$
- The slope of the asymptote after the breakpoint is ± 2 .
- If the damping is low, then the magnitude has a peak of height $1/(2\zeta)$ at the break point.
- The phase changes $\pm 180^\circ$ after the break point and attains $\pm 90^\circ$ at the breakpoint. The slope of the phase change depends on the damping coefficient and is abrupt for $\zeta = 0$.
- For a positive exponent, the peak at the breakpoint becomes a dip.



$\zeta = 0.1$: red, $\zeta = 0.5$: blue, $\zeta = 0.9$: green.

Short introduction to filters

A filter like $H(s)$ and $F(s)$ in the closed loop system are linear time invariant systems, like the process $G(s)$ or the controller $D(s)$.

Those filter are used for several reasons, two main reasons are:

- $F(s)$ reduces large control actions due to large abrupt reference changes.
- $H(s)$ reduces the effect of measurement noise on the closed loop system.

Note:

- Filter design is a research topic within Automatic Control and Signal Processing, and the topic is broad!
- This is just a short introduction how some simple but yet effective filters can be designed.

Aim: A filter should provide good tracking of a signal for a certain frequency range and should block the other frequencies in the signal.

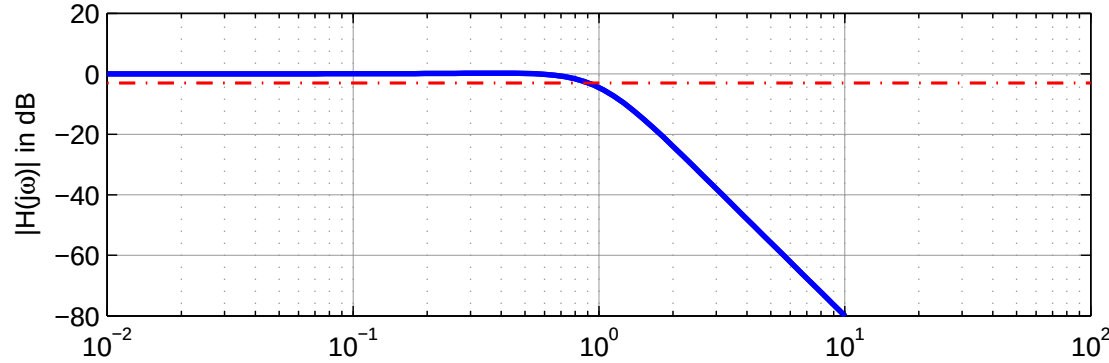
Three filter types:

- Low pass filter: Track frequencies below ω_0 and block frequencies above ω_0 .
- High pass filter: Track frequencies above ω_0 and block frequencies below ω_0 .
- Band pass filter: Track frequencies between ω_L and ω_H . Block all other frequencies.

Low pass filter

Specification: We want to track frequencies below ω_0 and block frequencies above that frequency.

Solution: Find a transfer function with magnitude 1 below ω_0 and magnitude 0 or very small above that frequency.



This filter provides a reduction of magnitude by the factor of 10000 over one decade of frequency!

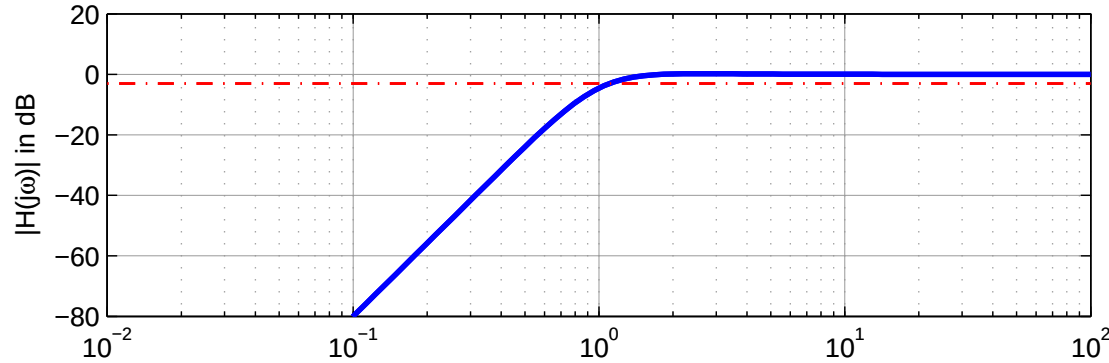
Design:

- Choose a first order transfer function with $\tau = 1/\omega_0$, or
- Choose a second order transfer function with $\omega_n = \omega_0$ and $\zeta \approx 0.9$.
- Adjust K_0 to get a magnitude of 1 below ω_0 .
- Put several of these filter in series if a faster reduction in magnitude is needed.

High pass filter

Specification: We want to track frequencies above ω_0 and block frequencies below that frequency.

Solution: Find a transfer function with magnitude 1 above ω_0 and magnitude 0 or very small below that frequency.



This filter provides a reduction of magnitude by the factor of 10000 over one decade of frequency!

Design:

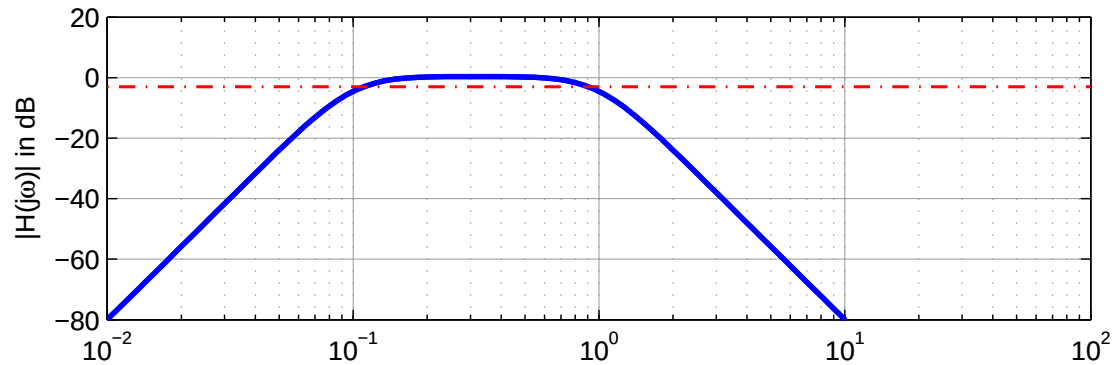
- Design a low pass filter.
- Multiply the term s^n to the numerator, where n is the degree of the denominator.
- Put several of these filter in series if a faster increase in magnitude is needed.

Band pass filter

Specification: We want to track frequencies in the region $\omega_L \dots \omega_H$ and block frequencies outside that region.

Solution: Find a transfer function which behaves like a high pass below ω_L and like a low pass above ω_H .

Thus, the band filter could be understood as the series of a low pass filter and a high pass filter.



Design:

- Design the low pass filter and high pass filter
- Combine the resulting transfer function as a series coupling.

Bode plot and stability

Remember:

- The stability of a closed loop system is determined by the roots of the characteristic polynomial $1 + L(s)$.
- In order for a test point to be a root of the characteristic equation, then the test point must fulfill the condition $1 + L(s) = 0$.

Frequency response:

When the frequency response is derived, the transfer function is evaluated for $j\omega$, with $\omega = -\infty \dots \infty$. Thus, the test points are all on the imaginary axis and a condition for poles on the stability margin can be formulated.

Condition for poles on the imaginary axis:

$$|L(j\omega)| = 1, \quad \angle L(j\omega) = -180^\circ$$

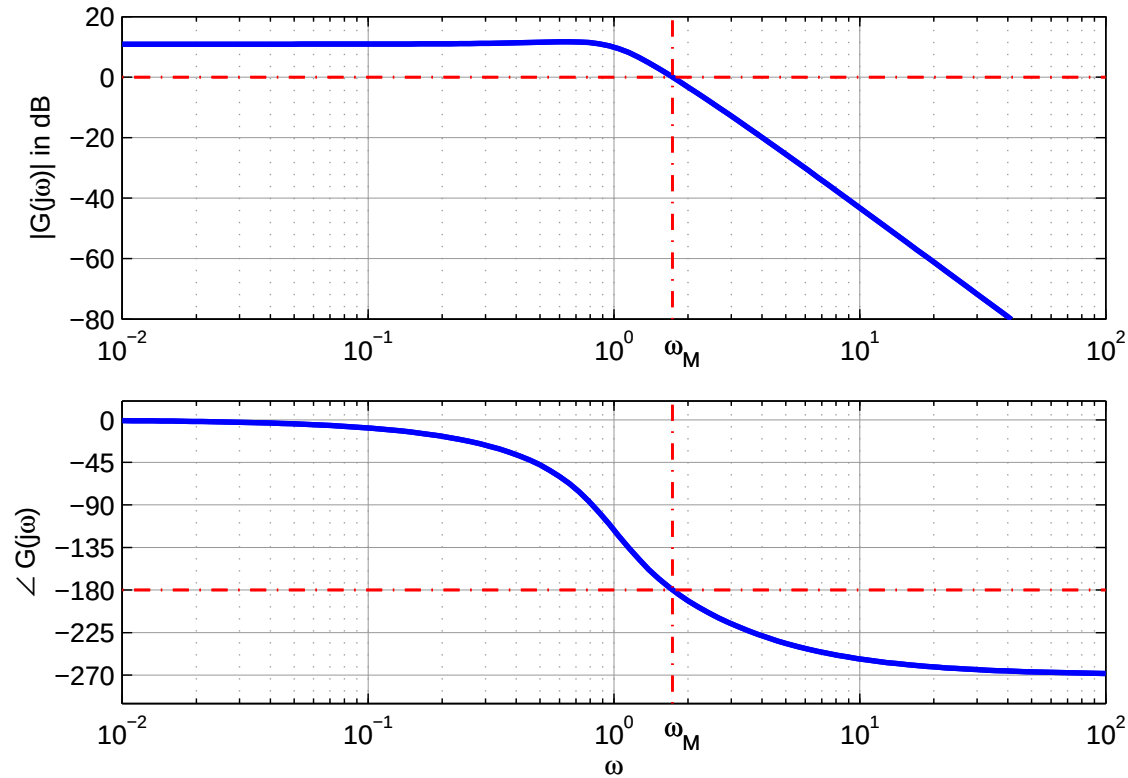
Systems that fulfill this condition are marginally stable.

Important notes:

- Using the open-loop transfer function $L(s)$, stability of the closed-loop system can be determined.
- The open-loop transfer function $L(s)$ has to be stable.

Example:

The closed loop system stability for a third order system shall be determined based on its open loop bode plot.



Conclusion: The system is marginally stable!

Bode plot and stability margins

Question: How can we quantify the distance to the stability border?

Idea: An open-loop system $L(s)$ that is stable can become unstable for an increase in gain when feedback is applied.

Stability condition: A closed loop system is stable if $L(s)$ is stable and

$$|L(j\omega)| < 1, \text{ where } \angle L(j\omega) = -180^\circ$$

Stability margins:

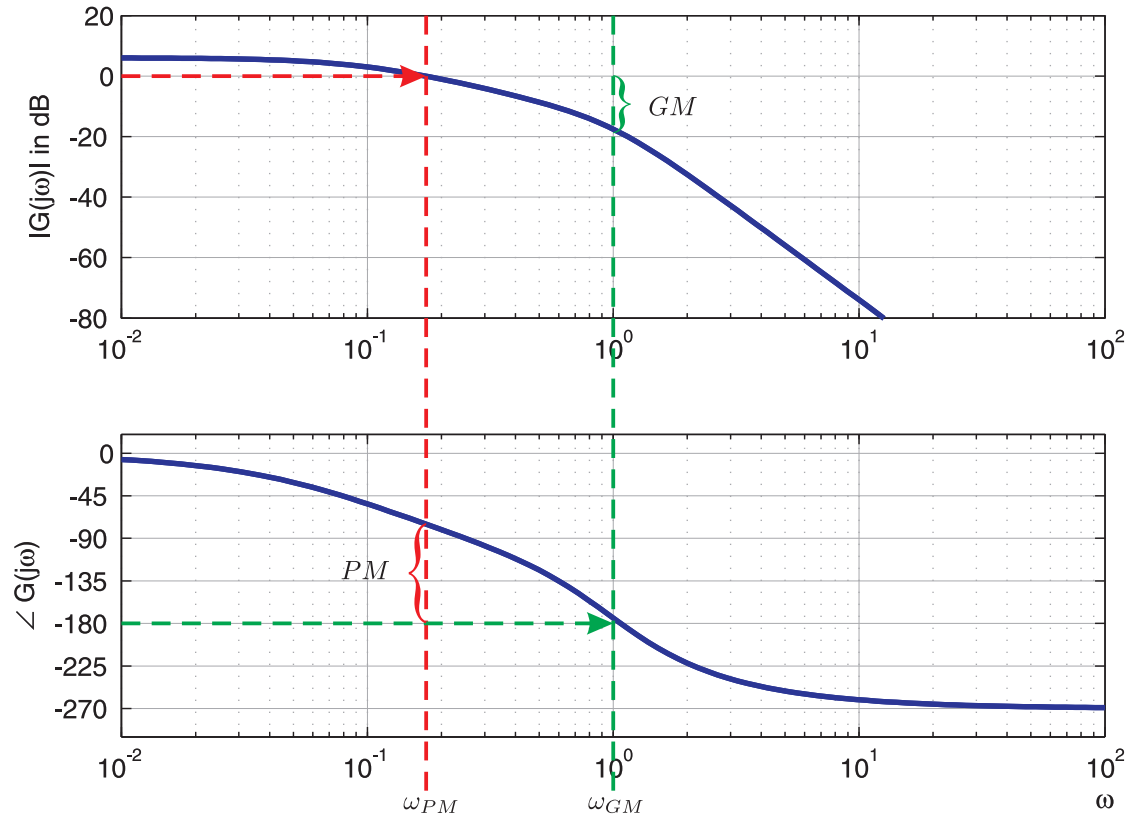
- Gain margin: This is the gain increase in $L(s)$ that would make the closed loop system marginally stable.
- Phase margin: This is the increase in negative phase shift in $L(s)$ that would make the closed loop system marginally stable.

How can we derive phase margin and gain margin?

1. Phase margin $PM = 180^\circ + \angle L(j\omega_{PM})$, with $\omega_{PM} = \omega|_{|L(j\omega)|=1}$

2. Gain margin $GM = \frac{1}{|L(j\omega_{GM})|}$, with $\omega_{GM} = \omega|_{\angle L(j\omega)=-180^\circ}$

Graphically: The phase and gain margin can also be directly assessed from the bode plot!



Question: What if $L(s)$ is unstable, or there are several points in frequency where $\angle L(j\omega) = -180^\circ$? $\Rightarrow$ Nyquist plot!

Time delays

Time delays can be represented in the time domain and Laplace domain as

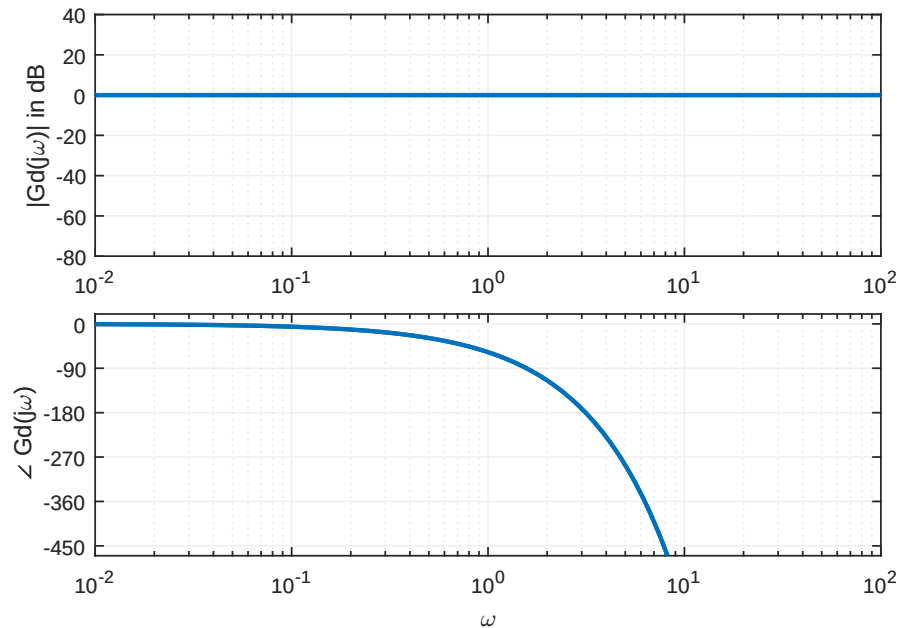
$$y(t) = u(t - T) \quad \Rightarrow \quad Y(s) = e^{-sT}U(s)$$

The frequency response of the term is now given by

$$G_d(j\omega) = e^{-j\omega T}$$

This is obviously the description of a vector in the complex plane with

$$|G_d(j\omega)| = 1, \quad \angle G_d(j\omega) = -\omega T$$



Observations:

- The magnitude is always one.
- The delay has a large negative phase contribution which increases linear in frequency.

Consequences:

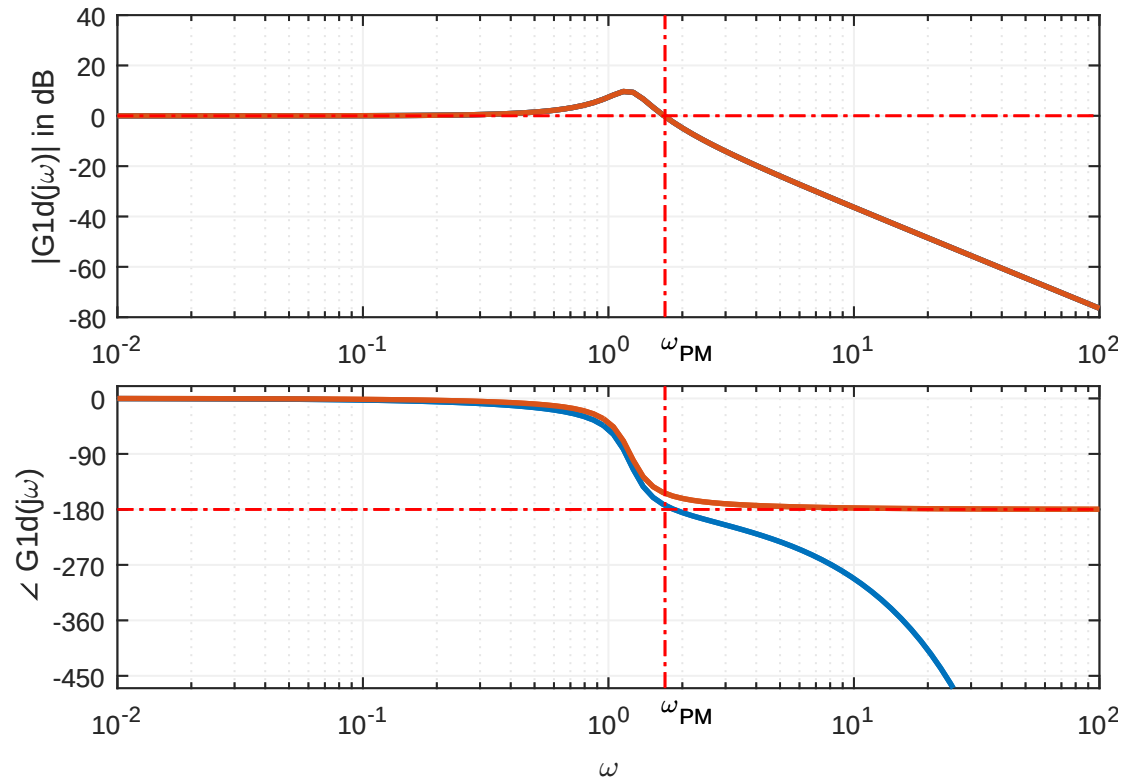
- The phase margin is reduced.
- This limits the achievable bandwidth for a closed loop system due to instability issues.

Let's have a look at the traverse crane example again with a time delay of $0.2s$:

$$G_1(s) = \frac{1.5}{s^2 + 0.4s + 1.5}$$

$$G_{1d}(s) = \frac{1.5e^{-0.2s}}{s^2 + 0.4s + 1.5}$$

We will now analyze the Bode plot for the systems.



Original system (green) and system with time delay (blue)

Notes:

- A closed loop system loses nearly all its phase margin.
- The closed loop system can not be made faster!

Additional material

There is a good video about Bode plots:

- Bode Plots, Introduction
https://www.youtube.com/watch?v=_eh1conN6YM

Then there are number of videos that explain how to create Bode plots in different cases:

- Bode plots by hand: Real constants
<https://www.youtube.com/watch?v=CSAp9ooQRT0>
- Bode plots by hand: Poles and zeros at the origin
<https://www.youtube.com/watch?v=E6R2XUEyRy0>
- Bode plots by hand: Real poles and zeros
https://www.youtube.com/watch?v=02Cw_4zd-aU
- Bode plots by hand: Complex poles and zeros
https://www.youtube.com/watch?v=GI1x9Yu__y8